February 1998

FAIRCHILD SEMICONDUCTOR™

74F245
Octal Bidirectional Transceiver with 3-STATE Outputs

General Description

The 'F245 contains eight non-inverting bidirectional buffers with 3-STATE outputs and is intended for bus-oriented applications. Current sinking capability is 24 mA (20 mA Mil) at the A ports and 64 mA (48 mA Mil) at the B ports. The Transmit/Receive ($T/\overline{R}$) input determines the direction of data flow through the bidirectional transceiver. Transmit (active HIGH) enables data from A ports to B ports; Receive (active LOW) enables data from B ports to A ports. The Output Enable input, when HIGH, disables both A and B ports by placing them in a High Z condition.

Features

- Non-inverting buffers
- Bidirectional data path
- A outputs sink 24 mA (20 mA Mil)
- B outputs sink 64 mA (48 mA Mil)
- Guaranteed 4000V minimum ESD protection

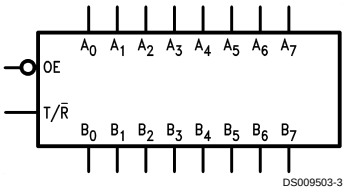
Ordering Code:

Commercial	Military	Package Number	Package Description
74F245PC		N20A	20-Lead (0.300" Wide) Molded Dual-In-Line
	54F245DM (Note 2)	J20A	20-Lead Ceramic Dual-In-Line
74F245SC (Note 1)		M20B	20-Lead (0.300" Wide) Molded Small Outline, JEDEC
74F245SJ (Note 1)		M20D	20-Lead (0.300" Wide) Molded Small Outline, EIAJ
74F245MSA (Note 1)		MSA20	20-Lead Molded Shrink Small Outline, EIAJ Type II
	54F245FM (Note 2)	W20A	20-Lead Cerpack
	54F245LM (Note 2)	E20A	20-Lead Ceramic Leadless Chip Carrier, Type C

Note 1: Devices also available in 13" reel. Use suffix = SCX, SJX and MSAX.

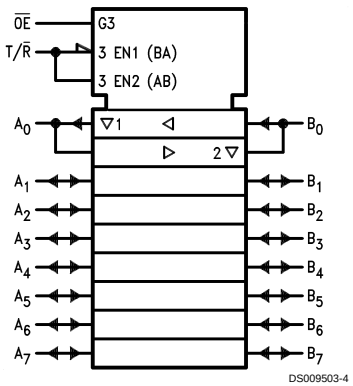
Note 2: Military grade device with environmental and burn-in processing. Use suffix = DMQB, FMQB and LMQB.

Logic Symbols



DS009503-3

IEEE/IEC

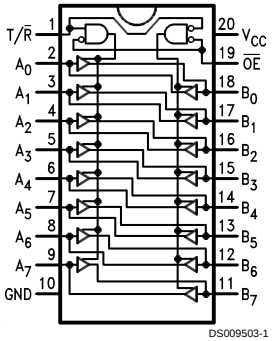


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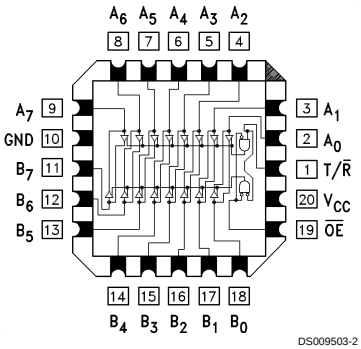
Connection Diagrams

Pin Assignment for DIP, SOIC, SSOP and Flatpak



DS009503-1

Pin Assignment for LCC



DS009503-2

Unit Loading/Fan Out

Pin Names	Description	54F/74F	
		U.L. HIGH/LOW	Input I_{IH}/I_{IL} Output I_{OH}/I_{OL}
$\overline{OE}$	Output Enable Input (Active LOW)	1.0/2.0	20 µA/−1.2 mA
T/$\overline{R}$	Transmit/Receive Input	1.0/2.0	20 µA/−1.2 mA
A_0–A_7	Side A Inputs or	3.5/1.083	70 µA/−0.65 mA
	3-STATE Outputs	150/40(38.3)	−3 mA/24 mA (20 mA)
B_0–B_7	Side B Inputs or	3.5/1.083	70 µA/−0.65 mA
	3-STATE Outputs	600/106.6(80)	−12 mA/64 mA (48 mA)

Truth Table

Inputs		Output
$\overline{OE}$	T/$\overline{R}$	
L	L	Bus B Data to Bus A
L	H	Bus A Data to Bus B
H	X	High Z State

H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial

Absolute Maximum Ratings (Note 3)

Storage Temperature	−65˚C to +150˚C
Ambient Temperature under Bias	−55˚C to +125˚C
Junction Temperature under Bias	−55˚C to +175˚C
Plastic	−55˚C to +150˚C
V_{CC} Pin Potential to Ground Pin	−0.5V to +7.0V
Input Voltage (Note 4)	−0.5V to +7.0V
Input Current (Note 4)	−30 mA to +5.0 mA
Voltage Applied to Output in HIGH State (with V_{CC} = 0V)	
Standard Output	−0.5V to V_{CC}
3-STATE Output	−0.5V to +5.5V
Current Applied to Output in LOW State (Max)	twice the rated I_{OL} (mA)
ESD Last Passing Voltage (Min)	4000V

Recommended Operating Conditions

Free Air Ambient Temperature	
Military	−55˚C to +125˚C
Commercial	0˚C to +70˚C
Supply Voltage	
Military	+4.5V to +5.5V
Commercial	+4.5V to +5.5V

Note 3: Absolute maximum ratings are values beyond which the device may be damaged or have its useful life impaired. Functional operation under these conditions is not implied.

Note 4: Either voltage limit or current limit is sufficient to protect inputs.

DC Electrical Characteristics

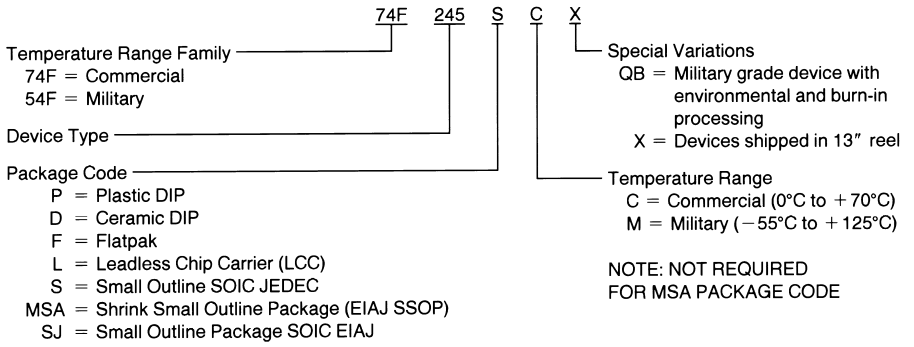
Symbol	Parameter		54F/74F Min	54F/74F Typ	54F/74F Max	Units	V_{CC}	Conditions
V_{IH}	Input HIGH Voltage		2.0			V		Recognized as a HIGH Signal
V_{IL}	Input LOW Voltage				0.8	V		Recognized as a LOW Signal
V_{CD}	Input Clamp Diode Voltage				−1.2	V	Min	I_{IN} = −18 mA
V_{OH}	Output HIGH Voltage	54F 10% V_{CC}	2.4					I_{OH} = −3 mA (A_n)
		54F 10% V_{CC}	2.0					I_{OH} = −12 mA (B_n)
		74F 10% V_{CC}	2.4			V	Min	I_{OH} = −3 mA (A_n)
		74F 10% V_{CC}	2.0					I_{OH} = −15 mA (B_n)
		74F 5% V_{CC}	2.7					I_{OH} = −3 mA (A_n)
V_{OL}	Output LOW Voltage	54F 10% V_{CC}			0.5			I_{OL} = 20 mA (A_n)
		54F 10% V_{CC}			0.55	V	Min	I_{OL} = 48 mA (B_n)
		74F 10% V_{CC}			0.5			I_{OL} = 24 mA (A_n)
		74F 10% V_{CC}			0.55			I_{OL} = 64 mA (B_n)
I_{IH}	Input HIGH Current	54F			20.0	µA	Max	V_{IN} = 2.7V
		74F			5.0			
I_{BVI}	Input HIGH Current Breakdown Test	54F			100	µA	Max	V_{IN} = 7.0V ($\overline{OE}$, T/$\overline{R}$)
		74F			7.0			
I_{BVIT}	Input HIGH Current Breakdown (I/O)	54F			1.0	mA	Max	V_{IN} = 5.5 V (A_n, B_n)
		74F			0.5			
I_{CEX}	Output HIGH Leakage Current	54F			250	µA	Max	V_{OUT} = V_{CC} (A_n, B_n)
		74F			50			
V_{ID}	Input Leakage Test	74F	4.75			V	0.0	I_{ID} = 1.9 µA All Other Pins Grounded
I_{OD}	Output Leakage Circuit Current	74F			3.75	µA	0.0	V_{IOD} = 150 mV All Other Pins Grounded
I_{IL}	Input LOW Current				−1.2	mA	Max	V_{IN} = 0.5V (T/$\overline{R}$, $\overline{OE}$)
I_{IH} + I_{OZH}	Output Leakage Current				70	µA	Max	V_{OUT} = 2.7V (A_n, B_n)
I_{IL} + I_{OZL}	Output Leakage Current				−650	µA	Max	V_{OUT} = 0.5V (A_n, B_n)
I_{OS}	Output Short-Circuit Current		−60		−150	mA	Max	V_{OUT} = 0V (A_n)
			−100		−225			V_{OUT} = 0V (B_n)
I_{ZZ}	Bus Drainage Test				500	µA	0.0V	V_{OUT} = 5.25V(A_n, B_n)
I_{CCH}	Power Supply Current			70	90	mA	Max	V_O = HIGH
I_{CCL}	Power Supply Current			95	120	mA	Max	V_O = LOW
I_{CCZ}	Power Supply Current			85	110	mA	Max	V_O = HIGH Z

AC Electrical Characteristics

Symbol	Parameter	74F $T_A = +25^\circ C$ $V_{CC} = +5.0V$ $C_L = 50$ pF			54F T_A, V_{CC} = Mil $C_L = 50$ pF		74F T_A, V_{CC} = Com $C_L = 50$ pF		Units
		Min	Typ	Max	Min	Max	Min	Max	
t_{PLH}	Propagation Delay	2.5	4.2	6.0	2.0	7.5	2.0	7.0	ns
t_{PHL}	A_n to B_n or B_n to A_n	2.5	4.2	6.0	2.0	7.5	2.0	7.0	
t_{PZH}	Output Enable Time	3.0	5.3	7.0	2.5	9.0	2.5	8.0	
t_{PZL}		3.5	6.0	8.0	3.0	10.0	3.0	9.0	ns
t_{PHZ}	Output Disable Time	2.0	5.0	6.5	2.0	9.0	2.0	7.5	
t_{PLZ}		2.0	5.0	6.5	2.0	10.0	2.0	7.5	

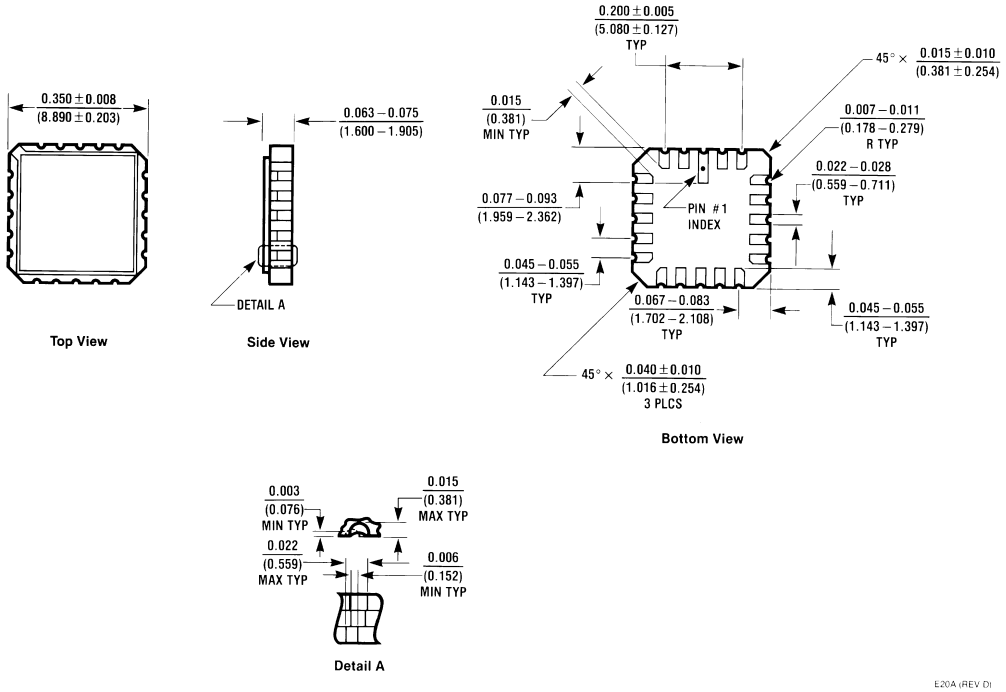
Ordering Information

The device number is used to form part of a simplified purchasing code where the package type and temperature range are defined as follows:

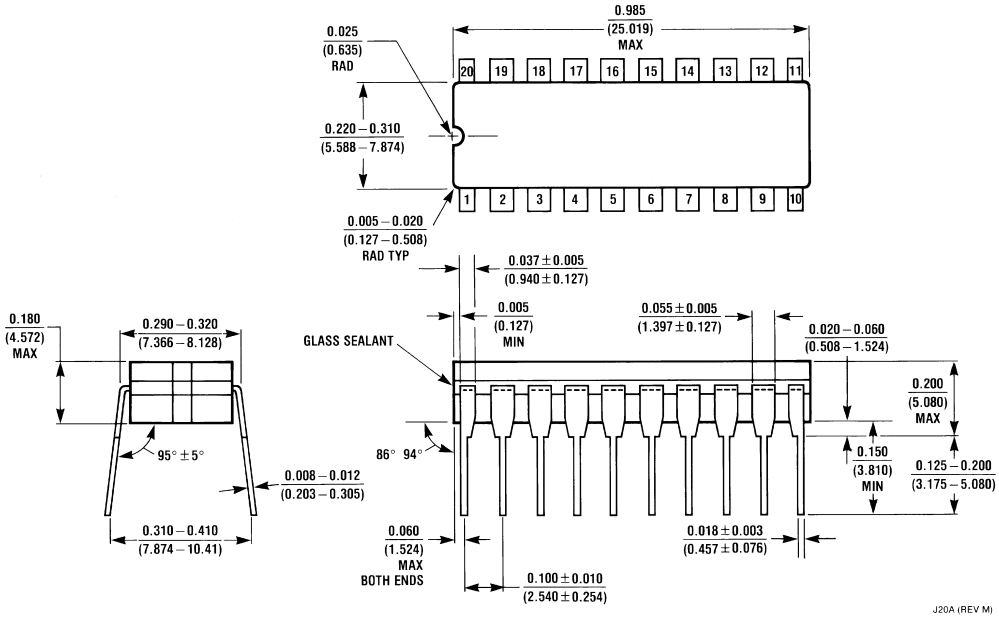


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Physical Dimensions inches (millimeters) unless otherwise noted

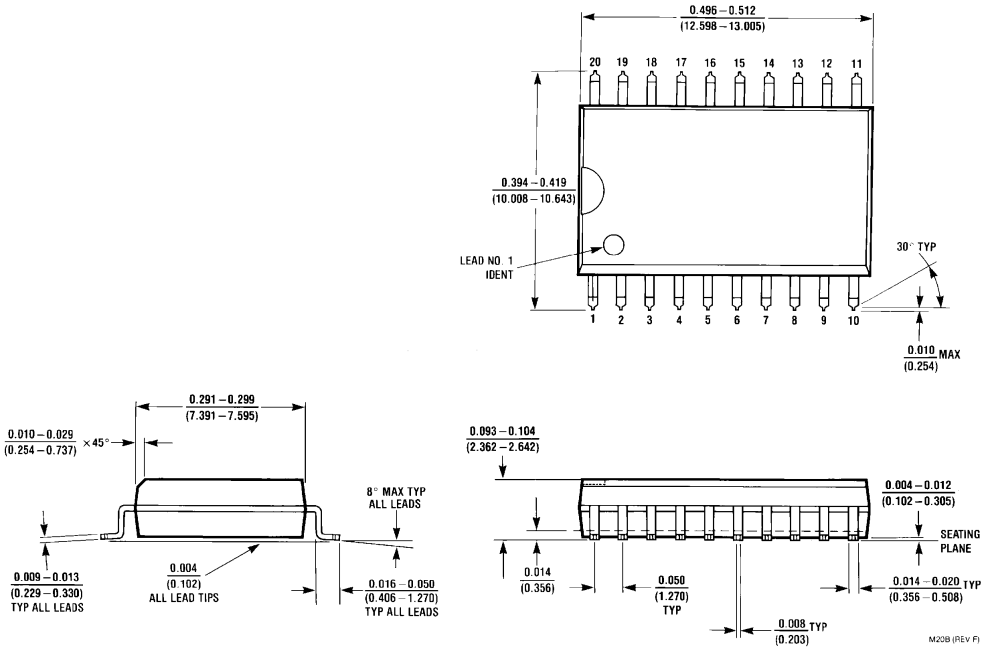


20-Lead Ceramic Leadless Chip Carrier (L)
Package Number E20A

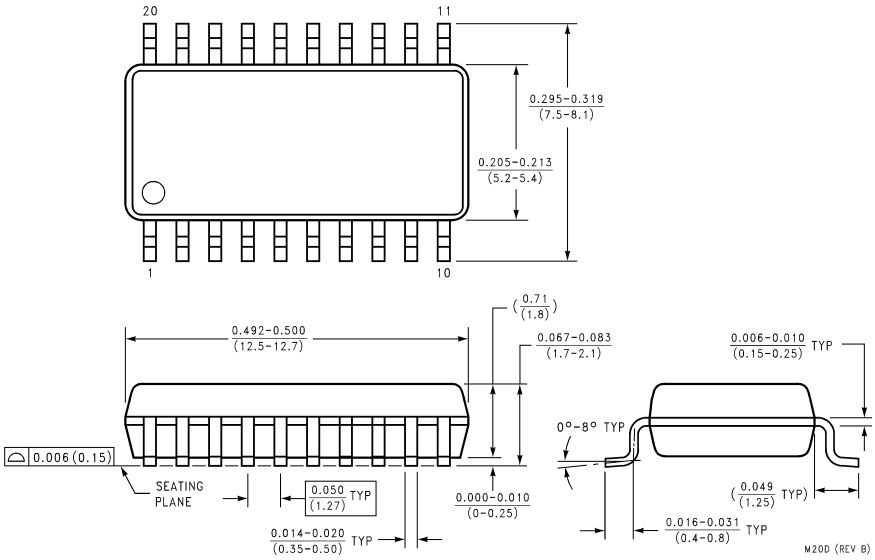


20-Lead Ceramic Dual-In-Line Package (D)
Package Number J20A

Physical Dimensions inches (millimeters) unless otherwise noted (Continued)

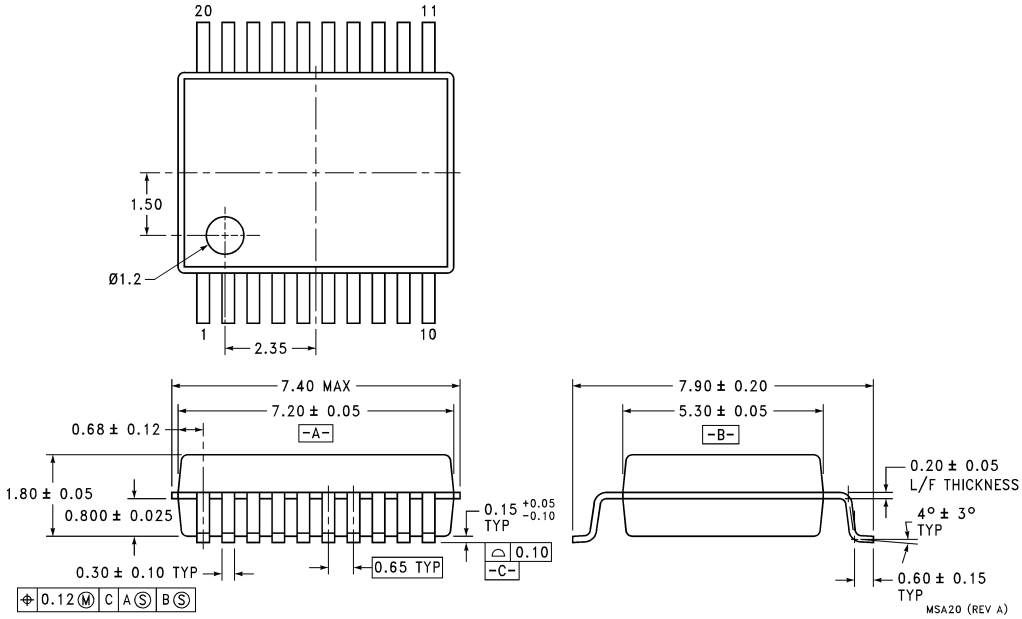


20-Lead (0.300" Wide) Molded Small Outline Package, JEDEC (S)
Package Number M20B

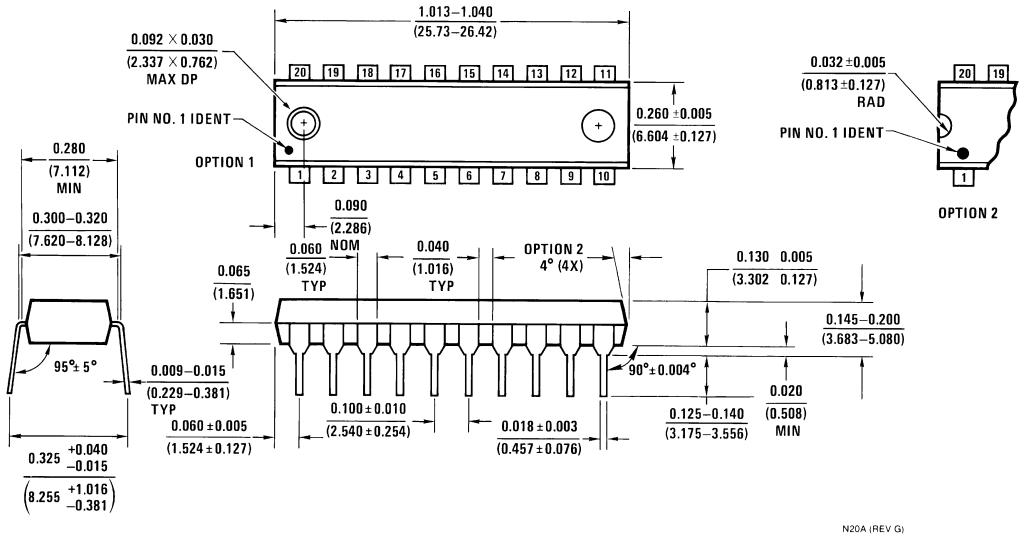


20-Lead (0.300" Wide) Molded Small Outline Package, EIAJ (SJ)
Package Number M20D

Physical Dimensions inches (millimeters) unless otherwise noted (Continued)

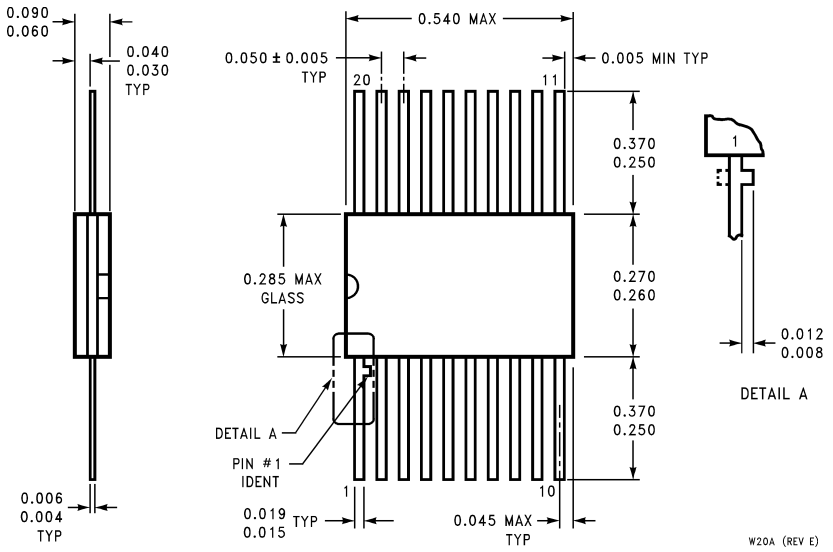


20-Lead Molded Shrink Small Outline Package, EIAJ, Type II (MSA)
Package Number MSA20



20-Lead Molded (0.300" Wide) Dual-In-Line Package (P)
Package Number N20A

Physical Dimensions inches (millimeters) unless otherwise noted (Continued)



20-Lead Ceramic Flatpak (F)
Package Number W20A

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